COEN 691 PERVASIVE HEALTHCARE – WINTER 2026

Assignment 1

AI Guidelines

- You may only use the AI provided in google colab to complete code for this assignment. Include all your prompts in your notebook.

Task description

You want to create a system for early detection of cognitive decline. As changes in daily activities are an indicator of cognitive decline, your first task is to recognize these activities from smartwatch data. You have collected data from one participant during several months and you will now create a model to recognize such activities.

Data description

The provided file contains the following columns:

- Stamp: the timestamp (year, month, day, hour, minute, seconds)
- user_acceleration_x: measured by a smartwatch accelerometer
- user_acceleration_y: measured by a smartwatch accelerometer
- user_acceleration_z: measured by a smartwatch accelerometer
- user_activity_label: the activity being performed at the time.

Instructions

Complete the following questions in a Google Colab notebook by clearly marking the question and answer in a text cell. Your answer may also consist of code, especially when a graph is required:

- We mentioned 5 types of applications in healthcare: wellness, monitoring of chronic disease, management of chronic disease, early detection, and behavior change. What type of healthcare application would you consider the intended system an example of? Explain
- Who is the intended user of the system?
- We have mentioned there are three components of a pervasive computing application: Sensing, thinking, and acting. Explain what each of them entails for the intended system.
- How many unique labels are there in the file?
- How much time is covered by the data? Are the time stamps continuous?
- What is the distribution of each activity in the file? (i.e. how many examples of each activity do we have)
- How do the acceleration values (x, y, z) vary across different activities?
- How does the frequency content of the acceleration signals vary across different activities (e.g., can you use Fourier analysis to identify distinctive frequency patterns)?
- How does the distribution of activities change over time (e.g., are some activities more common in the morning or evening)?

- Are there any temporal relationships between activities (e.g., do certain activities tend to occur in sequence or at specific times of day)?

Model questions (create a text cell with the answers):

- Are there any missing or duplicate values in the data? how do you handle them?
- How do you preprocess the data to remove noise or artifacts (e.g., filtering, normalization)?
- Are there any issues with data quality or consistency (e.g., variations in sampling rate, sensor calibration)?
- What model did you use (SVM, Random Forest, CNN, etc.)?
- Which features did you use?
- What is the size of the window you used for classification?

2. Create an activity recognition model using the provided file. Divide the data into training and testing and calculate the estimated f1-score using your test data. Then, use all the data to train a model. You will be evaluated based on the performance of your model on a hidden test data. You have been given an example test data file. Complete the code to calculate the f1-score on the hidden test data using the example test data file. We will change the test file for the complete file during the evaluation. The test file name is “df_test.csv”.

Submission

- Upload a .txt file with your name, student id, link to your google colab notebook. Share to p.lago.concordia@gmail.com and **ariana.olivier.ta@gmail.com**

- Download your notebook as .ipynb and upload it.